

Predicting the Trainability of Variational Quantum Circuits: A Data-Driven Model for Barren Plateaus

Md Habibur Rahman ¹  and Jaeho Kim ^{1,*} 

¹ Department of AI Convergence Engineering, Gyeongsang National University (GNU), Jinju, South Korea; habib@gnu.ac.kr (M.H.R.); jaeho.kim@gnu.ac.kr (J.K.)

* Correspondence: jaeho.kim@gnu.ac.kr; Tel.: +82-55-772-1373

Abstract

Barren plateaus—the exponential vanishing of cost-function gradients as a variational quantum circuit is scaled—are the central obstacle to training quantum machine-learning models, and diagnosing them today requires expensive gradient sampling on each candidate circuit. We reframe barren-plateau diagnosis as a supervised prediction problem: given only a circuit’s architecture (qubit count, depth, ansatz type, entanglement pattern, entangler gate, and cost-observable locality), can a classical model predict its trainability without sampling a single gradient? Using an exact statevector simulator we generate a dataset of 20,000 random variational circuits, each labelled by the empirical gradient variance $\text{Var}[\partial C / \partial \theta]$ estimated over 200 random parameter vectors via the parameter-shift rule. A gradient-boosted regressor predicts $\log_{10} \text{Var}$ with $R^2 = 0.685$ on held-out architectures, and a classifier separates barren from trainable circuits with area under the ROC curve 0.991. Crucially, in an extrapolation setting—training only on small circuits and testing on larger qubit counts that are progressively harder to simulate—the predictor retains $R^2 = 0.719$, indicating that the architectural signature of a barren plateau is learnable and transferable across scale. Permutation importance recovers established theory: cost-observable locality and entanglement structure dominate. We make no quantum-advantage claim; this is classical machine learning about quantum circuits, offering a cheap ansatz-screening heuristic and a data-driven lens on which design choices cause untrainability.

Keywords: barren plateaus; variational quantum circuits; quantum machine learning; trainability; gradient variance; statevector simulation; gradient boosting

1. Introduction

Variational quantum algorithms (VQAs) are the leading candidate for extracting value from noisy intermediate-scale quantum (NISQ) hardware [1]. A VQA encodes a problem into a parameterized quantum circuit (an *ansatz*), measures a cost function $C(\theta) = \langle \psi(\theta) | O | \psi(\theta) \rangle$, and updates the parameters θ with a classical optimizer. This template underlies variational quantum eigensolvers, quantum approximate optimization, and most quantum machine-learning (QML) classifiers [2–4].

The dominant obstacle to scaling this template is the *barren plateau* (BP) phenomenon: for broad classes of ansätze the variance of the cost gradient vanishes exponentially in the number of qubits, so the loss landscape becomes almost everywhere flat and gradient-based training stalls [5,6]. Barren plateaus are now known to be triggered by *global* cost observables [7], by highly expressive or deep ansätze [8], by poor initialization [9], and by

Received:

Accepted:

Published:

Copyright: © 2026 by the authors.

Submitted to *Entropy* for possible open access publication under the terms and conditions of the [Creative Commons Attribution \(CC BY\)](https://creativecommons.org/licenses/by/4.0/) license.

hardware noise [10]. Whether a given architecture suffers from a barren plateau is therefore a property of its *design*, not merely of a particular parameter setting.

The conventional way to check whether a candidate circuit is trainable is to *measure* it: sample many random parameter vectors, evaluate gradients, and estimate their variance. This is exactly the cost we would like to avoid when screening a large design space, and it becomes infeasible precisely in the large-qubit regime where barren plateaus matter most. This motivates our central question: *can a classical model predict a variational circuit's trainability directly from its architectural description, without sampling any gradients?*

We answer this empirically. We treat trainability prediction as a supervised learning problem over circuit *specs*. The features are cheap, integer-valued architectural descriptors; the label is the gradient variance, obtained once, offline, by exact statevector simulation. We then ask whether a model trained on this data (i) generalizes to unseen architectures, and (ii) *extrapolates* from small, easily simulated circuits to larger ones—the regime where a cheap predictor would be most useful.

The contributions of this article are as follows:

- We formulate barren-plateau diagnosis as a supervised prediction task: mapping a variational circuit's architecture to its gradient-variance trainability label, cast as both regression ($\log_{10} \text{Var}[\partial C / \partial \theta]$) and binary classification (barren vs. trainable).
- We release a reproducible pipeline that generates a labelled dataset of 20,000 random circuits using an exact, dependency-light statevector simulator, with fixed seeds and streaming output.
- We show a gradient-boosted model predicts trainability with $R^2 = 0.685$ on held-out architectures and separates barren circuits with AUC 0.991.
- We demonstrate extrapolation across scale: trained only on $n_q \leq 10$ qubits, the model still attains $R^2 = 0.719$ on larger, unseen circuit sizes.
- We recover the established theory through permutation feature importance (cost locality and entanglement structure dominate), giving the model an interpretable, physics-consistent basis.

We emphasise honest framing: we make *no* claim of quantum advantage. The model is classical, the data is classically simulated, and the contribution is a data-driven *diagnostic* for a known quantum-computing bottleneck [11,12].

2. Background and Related Work

2.1. Variational Quantum Circuits

A hardware-efficient ansatz [13] alternates layers of single-qubit rotations with a fixed pattern of two-qubit entangling gates. With n qubits and L layers the state is

$$|\psi(\theta)\rangle = \prod_{\ell=1}^L \left[U_{\text{ent}} \prod_{q=1}^n R_{a_{\ell,q}}(\theta_{\ell,q}) \right] |0\rangle^{\otimes n}, \quad (1)$$

where R_a is a Pauli rotation about axis $a \in \{x, y, z\}$ and U_{ent} is a layer of CZ or CX gates arranged linearly, circularly, or all-to-all. The cost is the expectation of a Pauli-Z string, $C(\theta) = \langle \psi(\theta) | O | \psi(\theta) \rangle$, with $O = Z_0$ (a *local* cost) or $O = Z_0 Z_1 \cdots Z_{n-1}$ (a *global* cost).

2.2. Barren Plateaus

A circuit family exhibits a barren plateau if $\text{Var}_{\theta}[\partial C / \partial \theta_k]$ decays exponentially in n [5]. Cerezo et al. [7] proved that *global* cost functions induce barren plateaus even at shallow depth, whereas *local* costs can remain trainable up to $O(\log n)$ depth—making cost locality a first-order design knob. Expressibility [8] and entanglement likewise correlate with flatter landscapes; structured architectures such as quantum convolutional networks can provably

avoid them [14]. A recent review consolidates these results [6]. This body of work is overwhelmingly *analytical*: it derives variance scaling laws for particular ansatz families. Our work is complementary and *empirical*—rather than proving a bound for one family, we learn a predictive map across a heterogeneous space of architectures.

2.3. The Parameter-Shift Rule

For a gate generated by an operator with two distinct eigenvalues, the exact gradient of the cost is obtained from two circuit evaluations [15,16]:

$$\frac{\partial C}{\partial \theta_k} = \frac{1}{2} [C(\theta + \frac{\pi}{2} e_k) - C(\theta - \frac{\pi}{2} e_k)]. \quad (2)$$

We use this rule to obtain exact per-sample gradients when estimating the variance label, so our labels are free of finite-difference bias.

2.4. Trainability, Simulability, and Honest Framing

Cerezo et al. [17] argue that the very structure guaranteeing absence of barren plateaus often also renders a model classically simulable. Bowles et al. [12] show that out-of-the-box classical models often match or beat QML models on standard tasks, and Schuld and Killoran [11] argue the field should de-emphasise “beating classical.” We adopt this stance: the object we predict is a property of quantum circuits, but the predictor and its value proposition are entirely classical. To our knowledge, casting barren-plateau severity as a directly *predicted* regression/classification target from cheap architectural features—and testing extrapolation across qubit count—has not been reported.

3. Materials and Methods

Figure 1 summarizes the full pipeline, from sampling a circuit spec to predicting its trainability.

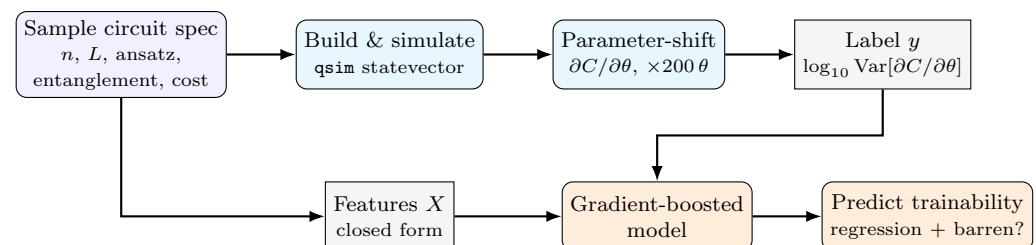


Figure 1. Method pipeline. A random circuit spec is built and simulated exactly with qsim; the parameter-shift gradient of the cost is sampled over 200 random parameter vectors to form the gradient-variance label y . Cheap closed-form architectural features X are read directly from the spec. A gradient-boosted model is trained on (X, y) to predict trainability.

3.1. Circuit Specification and Feature Space

Each datapoint is a *spec* that fully determines a circuit family. The feature vector handed to the model is listed in Table 1. All features are computed in closed form from the spec—no quantum simulation is needed to obtain them.

Table 1. Architectural features used to predict trainability. All are derived in closed form from the circuit spec.

Feature	Meaning
n_qubits	number of qubits n
n_layers	number of ansatz layers L
n_params	$n \times L$ trainable rotations
ansatz_type	fixed-ry vs. random-Pauli axes
entangle_pattern	linear / circular / all-to-all
entangler_gate	CZ vs. CX
cost_global	local Z_0 (0) vs. global $Z^{\otimes n}$ (1)
n_entanglers	entangling gates $\times$ layers
depth_ratio	L/n

3.2. Label Generation

For a spec, we draw 200 parameter vectors $\theta \sim \text{Unif}[0, 2\pi)^{nL}$, compute the exact parameter-shift gradient of the cost with respect to a fixed target rotation in the middle layer, and record the empirical variance. The regression label is $y = \log_{10} \max(\text{Var}[\partial C / \partial \theta], 10^{-18})$; the classification label is $\mathbb{I}[y < -2]$ (“barren”). Statevectors are evolved exactly with an in-house NumPy simulator, so labels reflect ideal (noiseless) trainability.

3.3. Dataset and Data Description

We sample 20,000 specs by drawing each design parameter independently and uniformly over the ranges in Table 2, with $n \in [2, 12]$ qubits and $L \in [1, 20]$ layers, each labelled with 200 gradient samples. Generation is embarrassingly parallel; every spec is seeded reproducibly from a root SeedSequence, and rows are streamed to CSV as they complete. The resulting dataset has 46.7% barren circuits and 50.1% global-cost circuits—a near-balanced classification target that avoids a degenerate majority baseline.

Table 2. Sampled design space. Each spec draws these parameters independently and uniformly.

Design parameter	Type	Sampled values
n_qubits	integer	2–12
n_layers	integer	1–20
ansatz_type	categorical (2)	fixed-ry, random-Pauli
entangle_pattern	categorical (3)	linear, circular, all-to-all
entangler_gate	categorical (2)	CZ, CX
cost_global	binary	local Z_0 , global $Z^{\otimes n}$

Table 3 summarizes the numeric features and the regression label. The design deliberately spans small, easily simulated circuits ($n = 2$) up to the $n = 12$ boundary of exact statevector simulation, and depths from a single layer to $L = 20$. The label $\log_{10} \text{Var}[\partial C / \partial \theta]$ ranges over roughly 18 orders of magnitude, with a floor at -18 where finite-sample variance underflows.

Table 3. Descriptive statistics of the numeric features and the trainability label over all 20,000 circuits.

Variable	Mean	Std	Min	25%	Median	Max
n_qubits	7.02	3.17	2.00	4.00	7.00	12.00
n_layers	10.48	5.80	1.00	5.00	11.00	20.00
n_params	73.56	55.55	2.00	28.00	60.00	240.00
n_entanglers	135.82	198.34	1.00	28.00	70.00	1320.00
depth_ratio	2.00	1.83	0.08	0.75	1.50	10.00
$\log_{10}$ Var (label)	-3.41	4.84	-18.00	-2.98	-1.85	-0.23

Crucially, the labels reproduce established barren-plateau theory directly from the data (Table 4). Global cost observables yield a mean $\log_{10}$ Var more than two orders of magnitude smaller than local costs (-4.49 vs. -2.33) and nearly double the barren fraction, exactly the cost-locality effect proved by Cerezo et al. [7]. Denser entanglement (all-to-all) and the CX entangler likewise push circuits toward barren plateaus. This structure is what makes the prediction task learnable and gives the trained model an interpretable, physics-consistent basis.

Table 4. Trainability broken down by design choice. Lower mean $\log_{10}$ Var and higher barren % indicate less trainable families. The data recovers known barren-plateau drivers.

Design choice	Count	Mean $\log_{10}$ Var	Barren %
<i>Cost-observable locality</i>			
local Z_0	9974	-2.33	30.9
global $Z^{\otimes n}$	10026	-4.49	62.4
<i>Entanglement pattern</i>			
linear	6649	-2.50	38.5
circular	6748	-2.96	48.8
all-to-all	6603	-4.79	52.8
<i>Entangler gate</i>			
CZ	10096	-2.31	38.1
CX	9904	-4.53	55.5
<i>Single-qubit rotation scheme</i>			
fixed-ry	9878	-2.80	42.0
random-Pauli	10122	-4.01	51.3

3.4. Models and Evaluation

To avoid privileging a single learner, we compare six model families spanning linear, instance-based, neural, and tree-ensemble inductive biases: a linear baseline (ridge regression / logistic regression), k -nearest neighbours, a multilayer perceptron (two hidden layers, 128–64), random forests, extremely randomized trees, and histogram-based gradient-boosted trees [18,19]. Features are standardized for the scale-sensitive models. We report two protocols for each: (i) a random hold-out split (20% test) measuring generalization to unseen architectures, and (ii) an *extrapolation* split, training on $n \leq 10$ and testing on all larger circuits—the headline test of whether small-circuit labels transfer to costlier regimes. All models use fixed random seeds so the reported numbers reproduce exactly. Interpretability is assessed with permutation feature importance on the best model. All code, command lines, and random seeds are released (Data Availability Statement), and the underlying simulator is available at <https://quantum.qbithabib.com>.

4. Results

4.1. Model Comparison

Table 5 compares all six model families on both tasks and both splits. Histogram-based gradient-boosted trees give the best held-out performance on both regression ($R^2 = 0.685$) and classification (AUC 0.991), so we adopt it as our model for the remainder of the paper. Two observations sharpen the picture. First, the linear baseline is far behind (regression $R^2 = 0.27$; classifier AUC 0.95), confirming that the map from architecture to trainability is genuinely nonlinear—simple rules of thumb do not capture it. Second, the neural network is competitive and in fact *edges out* boosting on the extrapolation regression ($R^2 = 0.75$ vs. 0.72), hinting that a learned representation transfers slightly better to larger circuits; we retain gradient boosting for its stronger in-distribution accuracy, interpretability, and training speed.

Table 5. Model comparison across both prediction tasks and both evaluation splits. Best hold-out result in **bold**.

Model	Regression R^2		Classification AUC	
	Hold-out	Extrap.	Hold-out	Extrap.
Hist Gradient Boosting	0.685	0.719	0.991	0.989
MLP	0.665	0.752	0.990	0.991
k -NN	0.645	0.639	0.986	0.979
Random Forest	0.640	0.716	0.985	0.988
Extra Trees	0.610	0.707	0.968	0.988
Linear baseline	0.274	0.297	0.953	0.889

4.2. Generalization to Unseen Architectures

On the random hold-out split, the regressor attains $R^2 = 0.685$ with mean absolute error 1.018 (in $\log_{10}$ units), and the classifier separates barren from trainable circuits with accuracy 0.950 and AUC 0.991. Figure 2 shows predicted versus true $\log_{10}$ gradient variance on the held-out set.

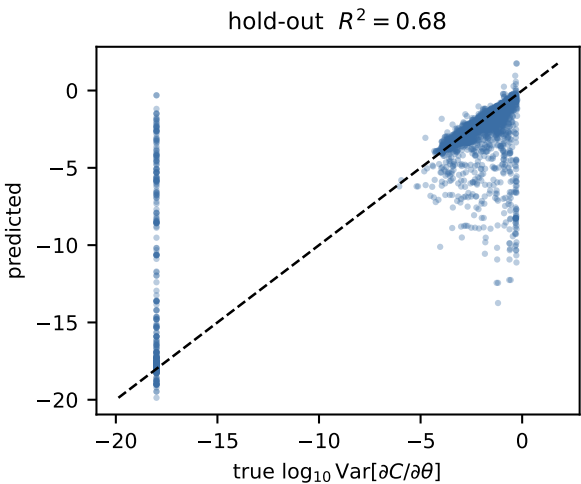


Figure 2. Predicted vs. true $\log_{10} \text{Var}[\partial C / \partial \theta]$ on held-out architectures. The dashed line is $y = x$.

4.3. Extrapolation across Qubit Count

Training only on $n \leq 10$ qubits (16 395 circuits) and testing on larger circuits (3 605 circuits), the regressor retains $R^2 = 0.719$ (MAE 1.264). Because the ground-truth cost of

labelling grows exponentially with qubit count, a predictor that transfers downward-in-cost is exactly what an ansatz-screening tool needs.

4.4. Trainability versus Qubit Count

Figure 3 shows the mean label as a function of qubit count, split by cost locality. Global-cost circuits exhibit the steep, monotone decay of gradient variance with qubit number that is the defining signature of a barren plateau, while local-cost circuits decay far more slowly. The dataset thus captures the qubit-scaling behaviour the predictor must learn to extrapolate.

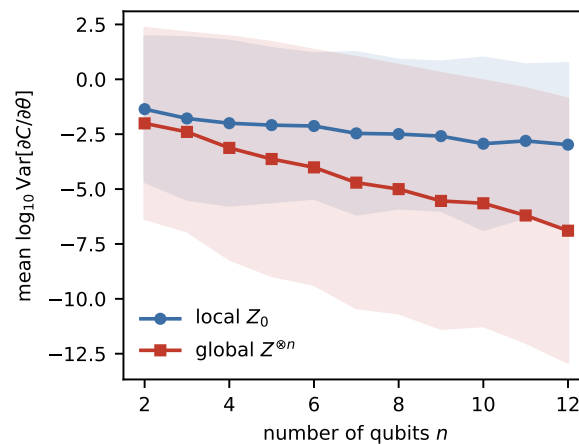


Figure 3. Mean $\log_{10} \text{Var}[\partial C / \partial \theta]$ vs. qubit count for local and global cost observables (shaded: ± 1 standard deviation). Global costs show the exponential decay characteristic of barren plateaus.

4.5. What Drives Trainability

Permutation feature importance (Figure 4) ranks cost-observable locality and entanglement structure at the top, consistent with theory: global costs [7] and denser entanglement [8] drive circuits toward barren plateaus. The model thus rediscovers, from data alone, the design rules the barren-plateau literature derived analytically.

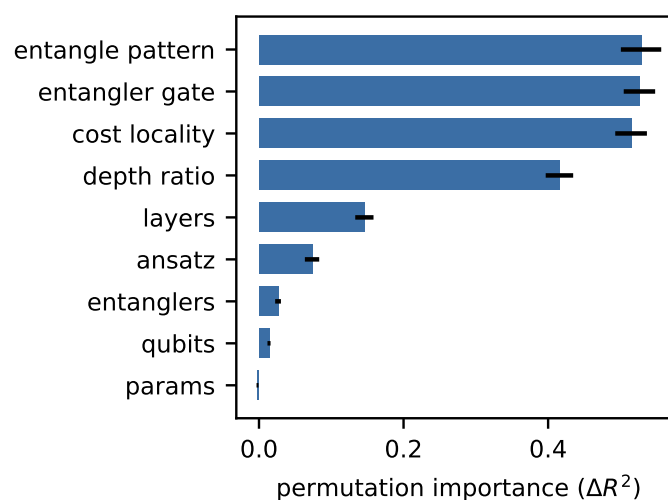


Figure 4. Permutation feature importance for the trainability regressor. Cost locality and entanglement structure dominate, matching barren-plateau theory [7,8].

5. Discussion

The predictor's accuracy and its physics-consistent feature ranking indicate that barren-plateau severity carries a strong, low-dimensional architectural signature—enough for a cheap classical model to recover without any gradient sampling. This suggests a practical workflow: screen a large space of candidate ansätze with the learned model, and reserve expensive gradient-variance estimation for the shortlist it flags as trainable.

Several limitations scope the claims. (i) Labels are limited to classically simulable qubit counts; the extrapolation result is evidence of transfer, not a proof. (ii) We use an ideal statevector simulator, so noise-induced plateaus [10] are out of scope. (iii) The spec space, while diverse, is a hardware-efficient family; other ansatz classes may need their own features. (iv) Gradient variance is estimated from finitely many samples, adding label noise at the trainable end. Seeds are fixed and reported, and the extrapolation split is defined purely by qubit count, so no large-circuit information leaks into training.

6. Conclusions

We recast barren-plateau diagnosis as supervised prediction and showed that a classical gradient-boosted model, trained on statevector-simulated labels, can predict a variational circuit's trainability from its architecture alone—and extrapolate across qubit count. The approach offers a cheap ansatz-screening heuristic and a data-driven confirmation of which design choices cause untrainability. Future work includes richer ansatz families, graph/sequence circuit encodings with graph-neural-network or transformer models, and incorporating a noise model to predict noise-induced plateaus.

Author Contributions: Conceptualization, methodology, software, formal analysis, data curation, visualization, and writing—original draft, M.H.R.; supervision, project administration, and writing—review and editing, J.K. Both authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: All code (statevector simulator, dataset generator, and training scripts), the exact command lines, and random seeds are released with this paper. The dataset is regenerable end-to-end with a single command. Interactive demonstrations of the underlying simulator are available at <https://quantum.qbithabib.com>.

Conflicts of Interest: The author declares no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

BP	Barren plateau
VQA	Variational quantum algorithm
QML	Quantum machine learning
NISQ	Noisy intermediate-scale quantum
MAE	Mean absolute error
AUC	Area under the ROC curve

References

1. Cerezo, M.; Arrasmith, A.; Babbush, R.; Benjamin, S.C.; Endo, S.; Fujii, K.; McClean, J.R.; Mitarai, K.; Yuan, X.; Cincio, L.; et al. Variational quantum algorithms. *Nature Reviews Physics* **2021**, *3*, 625–644. <https://doi.org/10.1038/s42254-021-00348-9>.
2. Havlíček, V.; Córcoles, A.D.; Temme, K.; Harrow, A.W.; Kandala, A.; Chow, J.M.; Gambetta, J.M. Supervised learning with quantum-enhanced feature spaces. *Nature* **2019**, *567*, 209–212. <https://doi.org/10.1038/s41586-019-0980-2>.
3. Farhi, E.; Neven, H. Classification with quantum neural networks on near term processors. *arXiv preprint arXiv:1802.06002* **2018**.
4. Schuld, M.; Bocharov, A.; Svore, K.M.; Wiebe, N. Circuit-centric quantum classifiers. *Physical Review A* **2020**, *101*, 032308. <https://doi.org/10.1103/PhysRevA.101.032308>.

5. McClean, J.R.; Boixo, S.; Smelyanskiy, V.N.; Babbush, R.; Neven, H. Barren plateaus in quantum neural network training landscapes. *Nature Communications* **2018**, *9*, 4812. <https://doi.org/10.1038/s41467-018-07090-4>. 219–220
6. Larocca, M.; Thanasilp, S.; Wang, S.; Sharma, K.; Biamonte, J.; Coles, P.J.; Cincio, L.; McClean, J.R.; Holmes, Z.; Cerezo, M. Barren plateaus in variational quantum computing. *Nature Reviews Physics* **2025**, *7*, 174–189. arXiv:2405.00781, <https://doi.org/10.1038/s42254-025-00813-9>. 221–223
7. Cerezo, M.; Sone, A.; Volkoff, T.; Cincio, L.; Coles, P.J. Cost function dependent barren plateaus in shallow parametrized quantum circuits. *Nature Communications* **2021**, *12*, 1791. <https://doi.org/10.1038/s41467-021-21728-w>. 224–225
8. Holmes, Z.; Sharma, K.; Cerezo, M.; Coles, P.J. Connecting ansatz expressibility to gradient magnitudes and barren plateaus. *PRX Quantum* **2022**, *3*, 010313. <https://doi.org/10.1103/PRXQuantum.3.010313>. 226–227
9. Grant, E.; Wossnig, L.; Ostaszewski, M.; Benedetti, M. An initialization strategy for addressing barren plateaus in parametrized quantum circuits. *Quantum* **2019**, *3*, 214. <https://doi.org/10.22331/q-2019-12-09-214>. 228–229
10. Wang, S.; Fontana, E.; Cerezo, M.; Sharma, K.; Sone, A.; Cincio, L.; Coles, P.J. Noise-induced barren plateaus in variational quantum algorithms. *Nature Communications* **2021**, *12*, 6961. <https://doi.org/10.1038/s41467-021-27045-6>. 230–231
11. Schuld, M.; Killoran, N. Is quantum advantage the right goal for quantum machine learning? *PRX Quantum* **2022**, *3*, 030101. <https://doi.org/10.1103/PRXQuantum.3.030101>. 232–233
12. Bowles, J.; Ahmed, S.; Schuld, M. Better than classical? The subtle art of benchmarking quantum machine learning models. *arXiv preprint arXiv:2403.07059* **2024**. 234–235
13. Kandala, A.; Mezzacapo, A.; Temme, K.; Takita, M.; Brink, M.; Chow, J.M.; Gambetta, J.M. Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets. *Nature* **2017**, *549*, 242–246. <https://doi.org/10.1038/nature23879>. 236–237
14. Pesah, A.; Cerezo, M.; Wang, S.; Volkoff, T.; Sornborger, A.T.; Coles, P.J. Absence of barren plateaus in quantum convolutional neural networks. *Physical Review X* **2021**, *11*, 041011. <https://doi.org/10.1103/PhysRevX.11.041011>. 238–239
15. Mitarai, K.; Negoro, M.; Kitagawa, M.; Fujii, K. Quantum circuit learning. *Physical Review A* **2018**, *98*, 032309. <https://doi.org/10.1103/PhysRevA.98.032309>. 240–241
16. Schuld, M.; Bergholm, V.; Gogolin, C.; Izaac, J.; Killoran, N. Evaluating analytic gradients on quantum hardware. *Physical Review A* **2019**, *99*, 032331. <https://doi.org/10.1103/PhysRevA.99.032331>. 242–243
17. Cerezo, M.; Larocca, M.; García-Martín, D.; et al. Does provable absence of barren plateaus imply classical simulability? *Nature Communications* **2025**, *16*, 7907. arXiv:2312.09121. 244–245
18. Ke, G.; Meng, Q.; Finley, T.; Wang, T.; Chen, W.; Ma, W.; Ye, Q.; Liu, T.Y. LightGBM: A highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems (NeurIPS)* **2017**, *30*. 246–247
19. Pedregosa, F.; Varoquaux, G.; Gramfort, A.; Michel, V.; Thirion, B.; Grisel, O.; et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research* **2011**, *12*, 2825–2830. 248–249

Disclaimer/Publisher’s Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content. 250–253